Polyline Editor Application — Analysis

Introduction

This document presents the Analysis Phase of the Polyline Editor application, following the interaction design process described in Chapter 5 of Human–Computer Interaction.

According to Section 5.1, interaction design focuses on developing systems that support user tasks effectively, rather than only implementing technical functionality. Therefore, this analysis emphasizes:

- Understanding user needs
- Identifying tasks and goals
- Defining interaction behavior
- Considering usability and system constraints

Understanding the Problem Domain

The system is a Polyline Drawing and Editing Application, where users:

- Create polylines using mouse clicks
- Edit shapes by moving or deleting vertices
- Perform actions using keyboard commands (b, d, m, etc.)

Key Observation

As described in Section 5.2, design is problem-solving under constraints.

Constraints in this system include:

- Limited input devices (mouse + keyboard)
- Real-time interaction requirement
- Need for continuous visual feedback

User Analysis (User-Centered Approach)

Target Users

- Students learning computer graphics
- Designers needing basic drawing tools
- General users with basic computer knowledge

User Characteristics

- Familiar with mouse interactions
- Expect intuitive controls (click, drag)
- Prefer visual and direct interaction

User Goals

- Quickly draw and modify shapes
- Easily edit polylines
- Recover from mistakes with minimal effort

Task Analysis

The system is broken down into tasks and subtasks following task-centered design.

Primary Tasks

1. Create polyline
2. Edit polyline
3. Manage drawing

Detailed Task Breakdown

➢ Create Polyline

- Press b
- Click multiple points
- System connects points with lines

➢ Move Vertex

- Select nearest point
- Drag to new position
- System updates connected segments

➢ Delete Vertex

- Select a point
- Remove it
- System reconnects adjacent points

➢ Refresh Display

- Redraw all polylines

- Quit Application
  - Exit safely

Interaction Analysis

Input Modalities

- Mouse → pointing, dragging
- Keyboard → command execution

Interaction Style

- Direct Manipulation (dragging points)
- Command-based interaction (keyboard shortcuts)

Interaction & Usability Considerations

- Direct manipulation improves learnability
- Keyboard commands improve efficiency
- System must provide immediate feedback after actions
- Important actions (like delete) should be clearly visible and controlled
- Selected vertex should be highlighted to avoid confusion

Constraints and Challenges

Technical Constraints

- Managing up to 100 polylines
- Efficient storage and updating of vertices

Interaction Constraints

- Difficulty in selecting precise points
- Mouse accuracy limitations

Usability Challenges

- Users may click wrong vertex
- Users may lose track of selected point
- Keyboard commands may not be intuitive initially

Conceptual Model (Mental Model Alignment)

Users naturally expect:

- Clicking → create point
- Dragging → move object
- Deleting → remove element

The system must align with this mental model to ensure:

- Better usability
- Reduced learning curve
- Fewer user errors

Design Implications (From Analysis)

Based on analysis, the system should ensure:

- Clear visualization of vertices and connections
- Immediate feedback for every interaction
- Simple and consistent controls
- Error tolerance (e.g., undo support, confirmation)

According to Section 5.3, design should be iterative, meaning user feedback must be used to refine the system.

Deliverables

As part of the analysis phase, the following deliverables are produced:

- User analysis and identification of target users
- Task decomposition and workflow understanding
- Interaction analysis (input methods and styles)
- Identification of system constraints and usability concerns
- Conceptual model alignment with user expectations

Lessons Learned

- Analysis is crucial before design and implementation
- User-centered thinking improves system usability
- Task analysis helps in simplifying complex systems
- Interaction design is not only about UI but also about user behavior

Confusions

- Whether keyboard-based commands are sufficient or GUI controls should be added
- Level of complexity required for additional features (e.g., undo, insert point)
- How detailed the analysis should be for academic vs real-world projects

Conclusion

The analysis phase identifies:

- User needs and behavior
- Core tasks and workflows
- Interaction patterns and constraints

Following HCI principles, this ensures that the Polyline Editor system will be:

- User-centered
- Task-oriented
- Efficient and usable